National University of Computer and Emerging Sciences

Software for Mobile Devices Sessional-I Exam (CS4039)

Date: March 1st, 2025

Course Instructor(s)

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Total Time: 1 hour
Total Marks: 30
Total Questions: 3

Roll No _____ Section _____ Student Signature _____

Do not write below this line

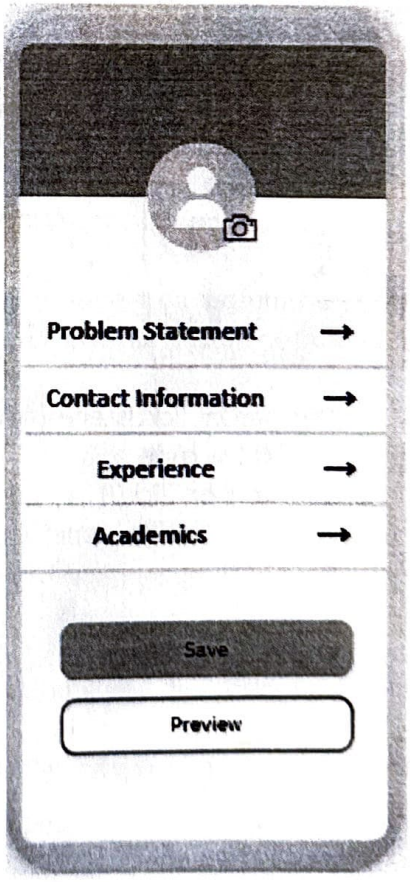
Attempt all questions on the answer sheet

CLO 1: Apply expertise of both frontend design and backend programming to create android mobile apps.

Question 1: **[10 marks]**

Layout of resume sharing app is attached. Profile image can be edited using icon at bottom right corner of profile image. Other features are editing the personal statement, contact information, experience and academic history.

You are required to write **xml code** to design this layout. Your proposed code should be responsive. No restrictions for color scheme. You can use colors of your own choice.



Question 2) Code Snippets [5+5+3 marks]

Write code snippets for the following tasks.

1. Write a Java code snippet to implement a fade-out animation on an ImageView when a button of id:"animBtn" is clicked. The animation should last for 2 seconds.
2. Write a Java code snippet to pass data of a user from MainActivity to SecondActivity using Intent. Assume the User has two fields: name (String) and age (int).
3. Write java code snippet to display **msg** to user using Toast. "Hello Mid 1 Exam" is stored in strings.xml and its name is "**msg**".

NOTE: Question 3 is different for different sections. Read the statement very carefully

Question 3 (For BCS-8A and BSE-8A Only): [7 marks]

In question 1 you have designed activity for resume sharing app. You are required to write necessary backend code in java for setting profile picture. Assume id of camera icon and profile picture in xml is "**ivCamera**" and "**ivProfilePic**". Write code to fulfill the following requirements:

- When user clicks on the camera icon, gallery should be opened
- User selects image from gallery and that image should be set as a profile pic.

Question 3 (For BCS-6A and 6B Only): [7 marks]

Suppose we want to design a Button class to replace Existing Java class in android. Class name is Button. To create its object we have to call its function called *getInstance()*

```
Button button=Button.getInstance(int id)
```

id will be an integer corresponding to its representation from xml layout file. Now we can call an event on button object named as *setOnClickListener*

```
button.setOnClickListener(new OnClickListener(){
        @Override
        void onClick(){
                //do something here.
        }
}
```

setOnClickListener() will take an object of OnClickListener.

You are required to propose architecture of Button class. You have to define the type and attributes of OnClickListener.